

Q1 a

Role of data processing and list three techniques prepare data for analysis

The role of data processing is to clean the data to generate a full dataset for analysis to maximise the information flow to user. Data processing is the **critical bridge** between raw data and meaningful insights. Its primary role is to transform **raw, unstructured, or messy data** into a **clean, structured, and reliable format** suitable for analysis, machine learning, or reporting.

1. Data cleaning, Identify missing values, to establish if there is a pattern, should the missing values be eliminated, replaced with median, mean or mode as appropriate.
2. Normalisation $(x_i - x_{\min}) / (x_{\max} - x_{\min})$ or $(x_i - \text{mean}(x)) / (\text{sd}(x))$
3. Feature selection **Data Reduction & Dimensionality Reduction**

Q1 b

Supervised learning involved labelled data with input and labelled output points. Supervised learning is a machine learning paradigm where an algorithm learns a mapping function from input variables (features) to an output variable (label) using a dataset of known, pre-labelled examples. **Classification: Goal:** Predict a **discrete category or class label**.

1. Decision Trees
2. Random forests
3. Gradient boosting trees

Decision trees are simple interpretable models, split data into subsets based on feature values producing tree like structures. **Decision Trees** are a non-parametric supervised learning method used for classification and regression. The goal is to create a model that predicts the value of a target variable by learning simple decision rules inferred from the data features. The tree is built by splitting the data into subsets based on the value of an input feature. This process is repeated on each derived subset in a recursive manner until the leaf nodes (where no further splitting is done) are reached.

Random forests is **ensemble method** that combines multiple decision trees to improve **predictive accuracy** and **control overfitting**. **Random Forests** is an ensemble learning method that operates by constructing a multitude of decision trees at training time and outputting the class that is the mode of the classes (classification) or mean prediction (regression) of the individual trees. Multiple trees are trained on different bootstrap samples (random samples with replacement) from the training data.

Gradient boosting trees build trees sequentially where new tree correct errors of previous tree, often leading to better performance but at cost of interpretability.

Sequential Building: Trees are built one at a time. Each new tree is fit to the residual errors (the difference between the current prediction and the true label) of the previous ensemble. **Gradient Descent:** The algorithm uses gradient descent to minimize a loss function (like mean squared error for regression, or log loss for classification) by adding trees that point in the negative gradient direction.

Key differences

Complexity Decision trees easy to interpret but Random forests and Gradient boosting trees more complex with better accuracy

Robustness Random forests are prone to overfitting compared to decision trees because of averaging of multiple trees.

Q1 c

Differentiate between overfitting and underfitting in the context of machine learning models. Provide examples of both.

Overfitting occurs when **model learns data too well** capturing noise and outliers with leads to poor generalisation of test data.

Example a complex polynomial such as a basis spline that fits every data point for interpolation and performs poorly when extrapolated to new data points.

Underfitting occurs when **model is too simple** to capture underlying patterns in data leading to **poor performance** on both training and test data sets. **Example 1: Linear Regression on Non-linear Data.** Fitting a straight line to clearly curved data (like a U-shape)

Q2 a

Discuss the concept of entropy used in decision trees to make splits.

Entropy is the **measure** of **impurity or disorder in dataset**. **Entropy** is a measure of **uncertainty, impurity, or disorder** in a dataset. In decision trees, it quantifies how "mixed" or "chaotic" the class distribution is at a node. The goal of splitting is to **reduce entropy**, creating purer child nodes. For a binary classification problem (two classes: positive/negative), entropy is calculated as:

$$\text{Entropy}(S) = -p^+ \times \log_2(p^+) - p^- \times \log_2(p^-)$$

Logarithm base 2 (giving entropy in "bits")

Calculation $\text{Entropy}(S) = -\sum p_i \log_2(p_i)$ where c is number of classes

Splitting criteria: Algorithm evaluate potential splits based on how much entropy is reduced after split in term of information gain. A split with higher information gain is preferred and results in purer child nodes.

Real-World Example: Medical Diagnosis

Problem: Diagnose disease (Yes/No) based on symptoms.

Initial State (Root Node):

- 100 patients: 60 healthy, 40 diseased
- Entropy = $-(0.6 \times \log_2 0.6 + 0.4 \times \log_2 0.4) \approx 0.971$

Q2 b

Discuss the concept of dimensionally reduction and explain two techniques for purpose.

Outline the main difference between wrapper and filter methods. **Filter Methods**

Concept:

Filter methods select features based on statistical measures of the features (like correlation with the target) without involving any machine learning model.

Process:

- Each feature is assigned a score based on a statistical test (like chi-square, ANOVA, correlation coefficient).
- Features are ranked by the score and either selected or removed.

Wrapper Methods

Concept:

Wrapper methods use a predictive model (like classifier or regressor) to evaluate the quality of feature subsets. They search for the best subset by training and evaluating the model on different feature combinations.

Process:

- A subset of features is selected.
- A model is trained on this subset.
- The model's performance is evaluated (using cross-validation to avoid overfitting).
- The process is repeated for different subsets

LIME (Local Interpretable Model=agnostic methods) generate locally faithful explanations by creating simple model around prediction of interest providing instance level of interpretation. LIME creates a local approximation of the model around a specific prediction by generating a new dataset of perturbed samples and the model's predictions on them. Then, it fits an interpretable model (like linear regression or decision tree) to this new dataset to explain the prediction.

SHAP (Shapley Additive Explanations) Provide global and local interpretability by calculation contribution of each features to prediction based on game theoretic Shapley values. SHAP is a **unified framework** for interpreting model predictions based on **Shapley values** from cooperative game theory. It explains the output of **any machine learning model** by assigning each feature an **importance value** for a particular prediction.

Every prediction is a "game" where features "cooperate" to produce the output. SHAP fairly distributes the "payout" (prediction difference from baseline) among the features according to their contributions.

Q2 c

TRUE/FALSE

Q3 Answer 4 out of next 5 points

(a) Time series

Concept a time and value data and stock market

ARIMA autoregressive iterative moving average

(b) Difference between local and global explainability

Local Explainability

Definition: Focuses on explaining **individual predictions**. It answers the question: *"Why did the model make this specific prediction for this particular instance?"*

Characteristics:

- **Instance-specific:** Explains one data point at a time.

Global Explainability

Definition: Focuses on explaining the **overall behavior of the model**. It answers the question: *"How does the model make predictions in general, across all instances?"*

Characteristics:

- **Model-wide:** Provides an understanding of the model's logic, important features, and decision boundaries.

© Bag of words and natural language processing

(d) Classification and regression techniques

(e) Role of neural networks

Neural networks are **computational models inspired by the human brain** that learn to recognize patterns and make decisions from data. They are the **foundation of modern deep learning** and have revolutionized AI by solving problems previously considered intractable.